

- 5 Four point charges, each of magnitude q , are placed at the corners of a square ABCD of side a as shown in Fig. 5.1 below.

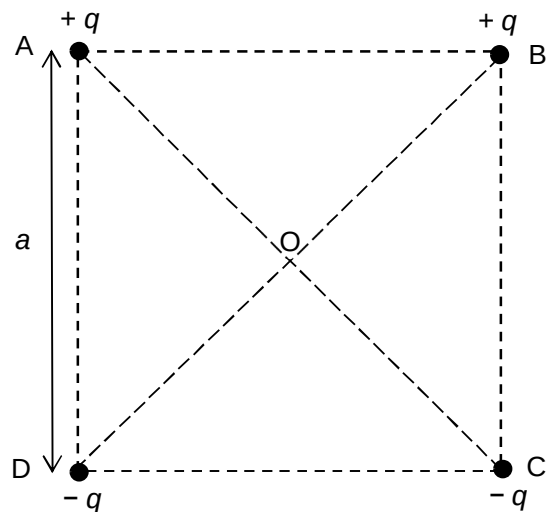


Fig. 5.1

Point O is the centre of the square.

- (a) Define *electric field strength* at a point.

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 [1]

- (b) (i) On Fig. 5.1, draw and label an arrow to show the resultant electric field strength E at O.

[1]

- (ii) Show that the electric field strength at O is $\frac{\sqrt{2}q}{\pi\epsilon_0 a^2}$.

[3]

- (c) A positively charged particle is projected into the square along a line that lies in the plane of the square as shown in Fig. 5.2. The line bisects the side AB.

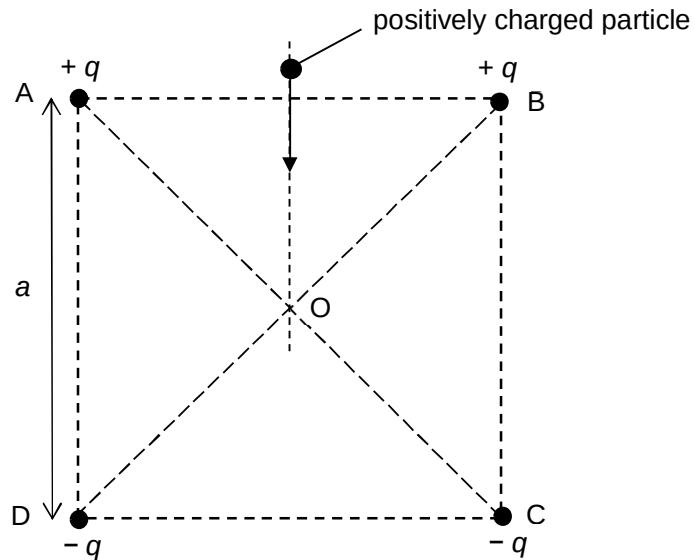


Fig. 5.2

Describe the motion of the particle after passing AB.

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 [2]

- (d) Determine in terms of q , a and ϵ_0 , the minimum kinetic energy required for the charge at C to move to infinity.

minimum kinetic energy = [3]

6 (a) A solenoid is connected to a variable d.c. supply as shown in Fig. 6.1